7+1 repository after the clean-up of 28 October 2026

The state of github.com/once-ere/7plus1 at commit 102f399 — everything that the spreadsheet 7plus1_status_2026-10-28.xlsx contains, with every formula it uses

Repository: github.com/once-ere/7plus1 — prepared for Nash
Commit described: 102f399 of 28 October 2026, 13:15 PDT (branch main)
Spreadsheet: 7plus1_status_2026-10-28.xlsx
Data file: tools/status_2026-10-28.json
Document generated by tools/make_docs.py on 2026-10-28

Abstract

On 28 October 2026 the documentation of the 7+1 project was completed with the auxiliary torus, the world brane and the radion (sections 25 and 26 of the notebook 7+1.nb), and every loose line of its 30 PDF documents was removed. This document records the state of the repository after that work, at commit 102f399: the 4 commits of that day up to it, the 30 documents with their 478 pages and their loose lines (40, then 12, now 0), the 16 paragraphs that were rewritten, with their formulas, the results of the twelve check programs (504 True, 0 False) and of the 479 checks of the notebook (479 True), the comparison of a fresh clone with the commit, and the private files that are kept out of the repository. It contains the six sheets of the spreadsheet 7plus1_status_2026-10-28.xlsx as tables and its 35 formulas with their values; every number comes from tools/status_2026-10-28.json, the data file of the spreadsheet; the values that LibreOffice computed and stored in the spreadsheet equal those computed here (35 of 35). The conventions of the 7+1 project, TeX's measure of a loose line and the functions of the spreadsheet are defined here; every equation used is written in this document.

Contents

1. Results at a glance **2**

2. Conventions **3**
2.1 The 7+1 spacetime . . . 3
2.2 Commits, times and counts . . . 3
2.3 Loose and overflowing lines . . . 3
2.4 The results of the check programs . . . 4

3. The commits of 28 October 2026 **4**

4. The 30 documents **5**
4.1 What each document is . . . 5
4.2 Loose lines, overflowing lines and errors . . . 5

5. The rewritten paragraphs **6**
5.1 Row 2, README: The Clifford products Θ^K, the sign σ_r and the lowered Θ_K . . . 7
5.2 Row 3, DIRAC16COMPLEX_PROVENANCE: The product and the left derivative of the Grassmann algebra . . . 7
5.3 Row 4, GRAVITY_PROVENANCE: The xAct definitions of the connection and the curvature tensors . . . 8
5.4 Row 5, STANDARD_MODEL_PROVENANCE: The mixing angle of two neutrino flavours in matter (MSW) . . . 8
5.5 Row 6, COUPLED_SOLUTIONS_PROVENANCE: The mode equation of pair creation . . . 8
5.6 Row 7, COUPLED_SOLUTIONS: The minisuperspace action and the identities of its variation 8
5.7 Row 8, COUPLED_SOLUTIONS_PROVENANCE: The comoving energy of one massless particle; the minisuperspace action and its identities . . . 9
5.8 Row 9, DARK_SECTOR: The Einstein-frame mass of a Kaluza–Klein level and its coupling to the radion . . . 9
5.9 Row 10, DIRAC16COMPLEX: The five momenta of the exact mode sum . . . 9

5.10 Row 11, DIRAC16COMPLEX_PROVENANCE: The orthogonal matrix U of the spinors . . 9
5.11 Row 14, KOHN_SHAM_GRAVITY_PROVENANCE: The completeness of the Θ_K and the Fierz identity . . . 9
5.12 Row 15, README_PROVENANCE: The notation of sections 7–10 . . . 10
5.13 Row 16, STANDARD_MODEL: The electroweak scale and the masses at tree level . . . 10
5.14 Row 17, STANDARD_MODEL_SYMBOLS: The matrix notation . . . 10

6. The checks **10**
6.1 The twelve check programs . . . 10
6.2 The notebook 7+1.nb . . . 11
6.3 A fresh clone of 102f399, rebuilt from scratch . . . 11

7. The private files **12**

8. The formulas of the spreadsheet **12**
8.1 Cells, ranges and functions . . . 12
8.2 The 35 formulas and their values . . . 12

9. How to reproduce **13**

1. Results at a glance

The first sheet of the spreadsheet, Summary. Every value is a formula that reads the other sheets (section 8); the last column names the sheet, and the section of this document that reproduces it.

item	value	where it comes from
Commit described (the latest when the spreadsheet was made)	102f399	Commits, last row (section 3)
Commits of 28 October 2026 up to the commit described	4	Commits (section 3)
File changes in these commits	206	Commits, total (a file changed twice counts twice) (section 3)
Documents (Markdown, LaTeX and PDF each)	30	Documents (section 4)
PDF pages	478	Documents, total (section 4)
Loose lines in the first build of 28 October	40	Documents, total (section 4)
Loose lines after commit 0654dcd	12	Documents, total (section 4)
Loose lines now	0	Documents, total (section 4)
Overflowing lines now	0	Documents, total (section 4)
LaTeX errors now	0	Documents, total (section 4)
Paragraphs rewritten (display, code block or rewording)	16	Fixes (section 5)
Check programs	12	Checks (section 6)
Check programs that exit with 0	12	Checks (section 6)
Results reported True by the programs	504	Checks, total (section 6)
Results reported False by the programs	0	Checks, total (section 6)
Checks in the notebook 7+1.nb	479	Checks, the notebook (section 6)
Notebook checks that returned True	479	Checks, the notebook (section 6)
Notebook message (error) cells	0	Checks, the notebook (section 6)
Markdown and LaTeX files identical after a fresh rebuild	60 of 60	Checks, fresh clone (section 6)
PDFs with the same text and page count after a fresh rebuild	30 of 30	Checks, fresh clone (section 6)
Private files on GitHub	0	Private files (section 7)

In words: the 4 commits of 28 October 2026 up to 102f399 made 206 file changes (a file changed in two commits counts twice). The 30 documents — each in Markdown, LaTeX and PDF — have 478 pages; the first full build of the day left 40 loose lines in them, 12 remained after commit 0654dcd, and none remain: no line is loose, none overflows, and LaTeX reports no error. For this, 16 paragraphs were rewritten. The 12 check programs all exit with code 0 and report 504 results True and 0 False; the 479 checks of the notebook 7+1.nb all returned True, with no message (error) cell. A fresh clone of 102f399, rebuilt from scratch, reproduces 60 of 60 Markdown and LaTeX files byte for byte and 30 of 30 PDFs with the same text and page count; no private file is on GitHub.

2. Conventions

2.1 The 7+1 spacetime

The 7+1 project constructs the octonions and the Clifford algebra of an eight-dimensional spacetime with the coordinates $X = (x_0, x_1, \ldots, x_7)$: x_0 is time, the direction of the octonion unit $e_0 = 1$; x_1, x_2, x_3 are ordinary space; $x_4, \ldots, x_7$ are four auxiliary space-like dimensions. Capital indices A, B run over $0, \ldots, 7$.

The signature: $\{7, 1, 0\}$ and $\mathrm{diag}(+1, -1, \ldots, -1)$. The 7+1 spacetime has one time-like dimension (x_0, the direction of the octonion unit $e_0 = 1$) and seven space-like dimensions (x_1, x_2, x_3 ordinary space, $x_4, \ldots, x_7$ auxiliary space). Two different objects describe this, and neither may be replaced by the other. (i) The tensor package xAct, used in section 12 of 7+1.nb exactly as in the author's notebook Einstein-Lovelock-7+1-Nash.nb, declares the metric with `DefMetric[{7, 1, 0}, g71[-a, -b], CD, …]`: the list $\{7, 1, 0\}$ declares seven space-like dimensions, one time-like dimension and no degenerate dimension — the numbers of entries $+1$ and -1 of $\eta_{71} = \mathrm{diag}(-1, +1, \ldots, +1)$ (`η71`), the same spacetime written in the sign convention of 4+4.nb (space-like = $+1$, time-like = -1). (ii) The matrix $\eta = -\eta_{71} = \mathrm{diag}(+1, -1, \ldots, -1)$ (`η17`) is simply the squares of the octonion basis: $e_0^2 = +1$ for the unit $e_0 = 1$ (time) and $e_i^2 = -1$ for the seven imaginary units $e_1, \ldots, e_7$ (space); more generally it is the real part of the product of two basis units, and it gives the quadratic form of the octonions:

$$\eta_{AB} = \mathrm{Re}(e_A e_B), \qquad e_0^2 = +1, \qquad e_i^2 = -1 \quad (i = 1, \ldots, 7),$$
$$\mathrm{Re}(x\,x) = x_0^2 - x_1^2 - \cdots - x_7^2 = \eta_{AB}\, x^A x^B \qquad (x = x^A e_A \in \mathbb{O}).$$

η gives the signs of the Clifford relation $\Gamma_A \Gamma_B + \Gamma_B \Gamma_A = 2\,\eta_{AB}\, 1_{16}$ of the real 16×16 Dirac matrices: time squares to $+1_{16}$ and each of the seven space directions to -1_{16}, exactly like the octonion units (in the notation $Cl(p, q)$, with p generators squaring to $+1$ and q to -1, the algebra $Cl(1, 7) \cong \mathbb{R}(16)$). It also gives the signs of the metric components: $g_{00} > 0$ along the time-like direction, negative squares along the space-like directions. Every component computed by 7+1.nb follows from the component matrix of the metric; the declared list fixes for xAct the sign of the determinant of the metric, `SignDetOfMetric[g71]` $= -1$, which is the same in both sign conventions, since $\det \eta = \det \eta_{71} = -1$. 7+1.nb checks `SignatureOfMetric[g71]` $= \{7, 1, 0\}$, `SignDetOfMetric[g71]` $= -1$, $\mathrm{Re}(e_A e_B) = \eta_{AB}$ and $\mathrm{Re}(x\,x) = \eta_{AB} x^A x^B$.

2.2 Commits, times and counts

- A commit is named by the first seven hexadecimal digits of its 40-digit SHA-1 object name in Git (`102f399` for `102f399d53720ba9b6bc05c6dbf8d70c68c9ac30`); section 3 lists the full names.
- Times are Pacific Daylight Time (PDT, UTC−7), the time zone of the author's computer, on which the commits were made; dates are written year-month-day.
- For a commit, "files changed", "lines added" and "lines removed" are the three numbers of `git show --shortstat`: a file counts once per commit that changes it, and a binary file (a PDF) counts as changed but adds and removes no lines.
- A document is one of the 30 documentation files of the repository at commit 102f399; each exists as Markdown (`.md`), LaTeX (`.tex`) and PDF (`.pdf`). Its "kind" is Main (a document), Symbols (its symbol dictionary) or Provenance (its provenance record); "covers" names the sections of 7+1.nb that it describes; its pages are those that `pdfinfo` reports.
- A document was "changed by" a commit if the commit changed its Markdown, LaTeX or PDF file.

2.3 Loose and overflowing lines

TeX sets each line of a paragraph in a box of the line width w. The material of the line has a natural width w_0, and its spaces can stretch by at most s (the total stretchability) or shrink by at most z (the total shrinkability) without becoming ugly. When a line must stretch by $t = w - w_0 > 0$, TeX measures how bad it is by its badness

$$b = \min\Big(10000,\ 100\Big(\frac{t}{s}\Big)^3\Big) \qquad (s > 0), \qquad b = 10000 \quad (s = 0,\ t > 0),$$

rounded to an integer. A LOOSE line — in the log of LaTeX the message `Underfull \hbox` — is a line whose badness exceeds 1000, the value of the parameter `\hbadness` that LaTeX sets: its spaces stretch to more than $10^{1/3} \approx 2.154$ times their stretchability,

$$\text{loose:} \quad b > 1000 \iff \frac{t}{s} > 10^{1/3} \approx 2.154.$$

An OVERFLOWING line — the message `Overfull \hbox` — is wider than the line even with all its spaces shrunk, by more than 0.1 pt, the value of the parameter `\hfuzz`:

$$\text{overflowing:} \quad w_0 - z - w > 0.1 \text{ pt}.$$

The numbers of loose and overflowing lines of a document are the numbers of the messages `Underfull \hbox` and `Overfull \hbox` in the log of its last LaTeX run (each message is one box: a line of a paragraph or of a table cell); a LaTeX error is a line of that log that begins with "!". The documents are compiled with `latexmk -xelatex`.

2.4 The results of the check programs

Each check program prints one line per result. The numbers of results reported True and False by a program P are

$$N_{\text{True}}(P) = \#\{\ell : \ \ell \text{ ends in ": True" or in "(True)"}\},$$
$$N_{\text{False}}(P) = \#\{\ell : \ \ell \text{ ends in ": False" or in "(False)"}\},$$

where ℓ runs over the lines of the output of P, # counts the elements of a set, and spaces between the colon and the word, and after the word, are allowed. The exit code is the status with which the program ends; 0 means success.

3. The commits of 28 October 2026

The sheet Commits. The four commits of 28 October 2026 up to 102f399, on the branch main of github.com/once-ere/7plus1:

commit	date and time (PDT)	files changed	lines added	lines removed
0654dcd	28 October 2026, 09:51	122	24,996	4,890
54862f0	28 October 2026, 10:15	12	66	19
93c500f	28 October 2026, 12:40	44	210	93
102f399	28 October 2026, 13:15	28	374	355
Total		**206**	**25,646**	**5,357**

What each commit contains:

- **0654dcd** (`0654dcddc090f5718ecd964c73dbda271f81757a`), 28 October 2026, 09:51 — *Sections 25-26: the auxiliary torus, the world brane, and the radion on the 4-torus and the 4-sphere.* Sections 25–26 of 7+1.nb and the `AUXILIARY_TORUS`, `WORLD_BRANE` and `RADION` documents (each with its symbols and provenance files).
- **54862f0** (`54862f028eb7d9a3a11b9076d79ff19cf0219f3f`), 28 October 2026, 10:15 — *Documentation tools: keep the evaluation record in `tools/eval_log_7p1.txt` for fresh-clone rebuilds.* The record of the evaluation of 7+1.nb, kept in the repository so that a fresh clone rebuilds the documents identically.
- **93c500f** (`93c500ff2ae198889d3d4abd4aafad0ea80f42de`), 28 October 2026, 12:40 — *Documentation: no loose lines left in any PDF; long file names and hashes may break.* The clean-up of the loose lines of the documents.
- **102f399** (`102f399d53720ba9b6bc05c6dbf8d70c68c9ac30`), 28 October 2026, 13:15 — *Documentation: count every check of `tools/ref_coupled.py` and `tools/ref_torus_brane.py`.* Corrected check counts (50 and 38 instead of 49 and 37), found while this spreadsheet was being prepared.

Commit 102f399 corrects two check counts that were found while the spreadsheet was being prepared; the spreadsheet and its three documents were added in the next commit.

4. The 30 documents

The sheet Documents, in two tables. The first gives what each document is; the second gives its loose lines in three builds — the first full build of the documents on 28 October 2026 (before commit 0654dcd), the rebuild of a fresh clone of 0654dcd, and the rebuild of a fresh clone of 102f399 ("now") — its overflowing lines and LaTeX errors now, and whether the commits 93c500f and 102f399 changed it.

4.1 What each document is

document	kind	covers (sections of 7+1.nb)	pages
README	Main	0–27 (the whole notebook)	32
README_SYMBOLS	Symbols	0–27 (the whole notebook)	13
README_PROVENANCE	Provenance	0–27 (the whole notebook)	66
DIRAC16COMPLEX	Main	7–10	22
DIRAC16COMPLEX_SYMBOLS	Symbols	7–10	15
DIRAC16COMPLEX_PROVENANCE	Provenance	7–10	27
GRAVITY	Main	11–14	19
GRAVITY_SYMBOLS	Symbols	11–14	8
GRAVITY_PROVENANCE	Provenance	11–14	15
KOHN_SHAM_GRAVITY	Main	15–16	9
KOHN_SHAM_GRAVITY_SYMBOLS	Symbols	15–16	5
KOHN_SHAM_GRAVITY_PROVENANCE	Provenance	15–16	8
STANDARD_MODEL	Main	17–19	11
STANDARD_MODEL_SYMBOLS	Symbols	17–19	6
STANDARD_MODEL_PROVENANCE	Provenance	17–19	8
COUPLED_SOLUTIONS	Main	20–23	19
COUPLED_SOLUTIONS_SYMBOLS	Symbols	20–23	10
COUPLED_SOLUTIONS_PROVENANCE	Provenance	20–23	16
DARK_SECTOR	Main	24	15
DARK_SECTOR_SYMBOLS	Symbols	24	8
DARK_SECTOR_PROVENANCE	Provenance	24	10
AUXILIARY_TORUS	Main	25, with 14–24 and 26	22
AUXILIARY_TORUS_SYMBOLS	Symbols	25, with 14–24 and 26	13
AUXILIARY_TORUS_PROVENANCE	Provenance	25, with 14–24 and 26	13
WORLD_BRANE	Main	25–26, with 17–24	24
WORLD_BRANE_SYMBOLS	Symbols	25–26, with 17–24	11
WORLD_BRANE_PROVENANCE	Provenance	25–26, with 17–24	13
RADION	Main	26	19
RADION_SYMBOLS	Symbols	26	9
RADION_PROVENANCE	Provenance	26	12
Total (30 documents)			**478**

4.2 Loose lines, overflowing lines and errors

document	loose, first build	loose after 0654dcd	loose now	overflowing now	errors now	changed by 93c500f	changed by 102f399
README	4	0	0	0	0	no	yes
README_SYMBOLS	0	0	0	0	0	no	no
README_PROVENANCE	11	1	0	0	0	yes	yes
DIRAC16COMPLEX	2	1	0	0	0	yes	no
DIRAC16COMPLEX_SYMBOLS	0	0	0	0	0	no	no
DIRAC16COMPLEX_PROVENANCE	6	1	0	0	0	yes	yes
GRAVITY	1	1	0	0	0	yes	no
GRAVITY_SYMBOLS	0	0	0	0	0	no	no
GRAVITY_PROVENANCE	4	1	0	0	0	yes	no
KOHN_SHAM_GRAVITY	0	0	0	0	0	no	no
KOHN_SHAM_GRAVITY_SYMBOLS	0	0	0	0	0	no	no

document	loose, first build	loose after 0654dcd	loose now	overflowing now	errors now	changed by 93c500f	changed by 102f399
KOHN_SHAM_GRAVITY_PROVENANCE	1	1	0	0	0	yes	no
STANDARD_MODEL	1	1	0	0	0	yes	no
STANDARD_MODEL_SYMBOLS	1	1	0	0	0	yes	no
STANDARD_MODEL_PROVENANCE	1	0	0	0	0	no	no
COUPLED_SOLUTIONS	2	1	0	0	0	yes	yes
COUPLED_SOLUTIONS_SYMBOLS	0	0	0	0	0	no	no
COUPLED_SOLUTIONS_PROVENANCE	3	2	0	0	0	yes	yes
DARK_SECTOR	1	1	0	0	0	yes	no
DARK_SECTOR_SYMBOLS	0	0	0	0	0	no	no
DARK_SECTOR_PROVENANCE	2	0	0	0	0	no	no
AUXILIARY_TORUS	0	0	0	0	0	no	yes
AUXILIARY_TORUS_SYMBOLS	0	0	0	0	0	no	no
AUXILIARY_TORUS_PROVENANCE	0	0	0	0	0	no	yes
WORLD_BRANE	0	0	0	0	0	no	yes
WORLD_BRANE_SYMBOLS	0	0	0	0	0	no	no
WORLD_BRANE_PROVENANCE	0	0	0	0	0	no	yes
RADION	0	0	0	0	0	no	no
RADION_SYMBOLS	0	0	0	0	0	no	no
RADION_PROVENANCE	0	0	0	0	0	no	no
Total	**40**	**12**	**0**	**0**	**0**	**11 yes**	**9 yes**

The loose lines of the first build were all in the README and in the documents of the earlier stages; the documents of the auxiliary torus, the world brane and the radion had none. Commit 0654dcd let code break inside long file names and hashes, set the headings ragged right and rewrote 5 paragraphs, which removed 28 of the 40 loose lines; commit 93c500f rewrote 11 more paragraphs, which removed the other 12 (section 5).

5. The rewritten paragraphs

The sheet Fixes. "How": Line breaking — a change of the typesetting rules; Display — a long inline formula moved into a displayed equation; Code block — a long piece of code moved into a block of its own; Reworded — the sentence rewritten so that TeX can break its lines evenly; Correction — a correction of content. Every rewritten paragraph keeps its content.

#	document	section of the document	how	change	commit
1	all 30 documents	—	Line breaking	Code may break after `/`, `_` and `\`, long hashes every 8 characters; headings are set ragged right	0654dcd
2	README	12.4 The nine-channel self-coupling and Kohn–Sham theory in gravitational fields	Display	The definitions of Θ^K, σ_r and Θ_K	0654dcd
3	DIRAC16COMPLEX_PROVENANCE	3.4 Grassmann algebra, quantization and the canonical anticommutators	Display	The product and the left derivative of the Grassmann algebra	0654dcd
4	GRAVITY_PROVENANCE	2.2 The reference notebook Einstein-Lovelock-7+1-Nash.nb	Code block	The xAct definitions `CD = LC[g]`, `RiemannCD`, `RicciCD`, `RicciScalarCD`, `EinsteinCD`	0654dcd
5	STANDARD_MODEL_PROVENANCE	4. Derivations	Display	The mixing angle in matter (MSW)	0654dcd
6	COUPLED_SOLUTIONS_PROVENANCE	4. Derivations	Display	The mode equation of pair creation	0654dcd
7	COUPLED_SOLUTIONS	4. The coupled field equations and the energy–momentum tensors	Display	The minisuperspace action and the identities of its variation	93c500f

#	document	section of the document	how	change	commit
8	COUPLED_SOLUTIONS_PROVENANCE	4. Derivations	Display	The comoving energy of one massless particle; the minisuperspace action and its identities	93c500f
9	DARK_SECTOR	6.1 From the solutions of regimes I and II: overproduction	Display	The Einstein-frame mass of a Kaluza–Klein level and its coupling to the radion, $g_n = \lvert dM_E/d\phi \rvert$	93c500f
10	DIRAC16COMPLEX	6.2 Mode expansion and canonical anticommutators	Display	The five momenta of the exact mode sum	93c500f
11	DIRAC16COMPLEX_PROVENANCE	3.3 Real plane-wave spinors	Display	The orthogonal matrix U of the spinors and $U^{\mathsf{T}}hU$	93c500f
12	GRAVITY	9. How to reproduce	Reworded	The generator command and its files	93c500f
13	GRAVITY_PROVENANCE	1. The task	Reworded	The item on the frame matrices and the spin connection	93c500f
14	KOHN_SHAM_GRAVITY_PROVENANCE	4. Derivations	Display	The completeness of the Θ_K and the Fierz identity	93c500f
15	README_PROVENANCE	6. The second stage: sections 7–10 of 7+1.nb	Display	The notation of sections 7–10	93c500f
16	STANDARD_MODEL	1. Results at a glance	Reworded	The electroweak scale and the tree-level masses	93c500f
17	STANDARD_MODEL_SYMBOLS	1. Conventions	Reworded	The matrix notation	93c500f
18	8 documents: README, README_PROVENANCE, COUPLED_SOLUTIONS, AUXILIARY_TORUS and WORLD_BRANE, and the provenance files of the last three	—	Correction	Every check of tools/ref_coupled.py (50) and tools/ref_torus_brane.py (38) is now counted and listed; before, 49 and 37. A ninth document, DIRAC16COMPLEX_PROVENANCE, changed only in the SHA-256 of tools/doc_stage3_common.py that it records	102f399

The formulas of the rewritten paragraphs, exactly as the documents write them (a final comma, colon or semicolon that continues the sentence of the document is written here as a full stop). Rows 1, 12, 13 and 18 contain no formula.

5.1 Row 2, README: The Clifford products Θ^K, the sign σ_r and the lowered Θ_K

For the 256 increasing multi-indices $K = (A_1 < \cdots < A_r)$, $r = 0, \ldots, 8$, of the eight real 16×16 Dirac matrices Γ^A:

$$\Theta^K = i^{\sigma_r}\Gamma^{A_1}\cdots\Gamma^{A_r}, \qquad \sigma_r = \frac{r(r-1)}{2} \bmod 2, \qquad \Theta_K = \eta_{A_1A_1}\cdots\eta_{A_rA_r}\Theta^K.$$

The factor i^{σ_r} makes each $\bar{\Psi}\Theta^K\Psi$ real; η_{AA} are the diagonal entries of $\eta = \mathrm{diag}(+1, -1, \ldots, -1)$; for $r = 0$ the product is 1_{16}.

5.2 Row 3, DIRAC16COMPLEX_PROVENANCE: The product and the left derivative of the Grassmann algebra

For increasing index sets K, K' of the anticommuting generators θ_k ($\theta_k\theta_l = -\theta_l\theta_k$), with θ_K the product of the generators of K in increasing order:

$$\theta_K\theta_{K'} = \mathrm{sign}(\pi)\,\theta_{\mathrm{sort}(K\cup K')} \qquad (\text{zero if } K\cap K' \neq \emptyset),$$
$$\frac{\partial^L(\theta_{k_1}\cdots\theta_{k_r})}{\partial\theta_{k_j}} = (-1)^{j-1}\,\theta_{k_1}\cdots\widehat{\theta_{k_j}}\cdots\theta_{k_r}.$$

π is the permutation that sorts the sequence K followed by K', $\mathrm{sort}(K\cup K')$ the sorted union, ∂^L the derivative from the left, and the hat marks the factor that is left out.

5.3 Row 4, GRAVITY_PROVENANCE: The xAct definitions of the connection and the curvature tensors

As the author's notebook Einstein-Lovelock-7+1-Nash.nb writes them, after a component metric g has been assigned (Wolfram Language with the packages xAct, xCoba and xTras):

```
CD = LC[g]; RiemannCD = Riemann[CD]; RicciCD = Ricci[CD];
RicciScalarCD = RicciScalar[CD]; EinsteinCD = Einstein[CD];
```

After `CD = LC[g]` the abstract covariant derivative `CD` is the Levi-Civita connection of the component metric g, and `RiemannCD`, `RicciCD`, `RicciScalarCD`, `EinsteinCD` are its Riemann tensor, Ricci tensor, scalar curvature and Einstein tensor.

5.4 Row 5, STANDARD_MODEL_PROVENANCE: The mixing angle of two neutrino flavours in matter (MSW)

Diagonalizing the two-flavour Hamiltonian in matter gives the mixing angle in matter θ_m:

$$\frac{\Delta m^2}{4E}\begin{pmatrix} -\cos 2\theta & \sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{pmatrix} + \mathrm{diag}(V,0) \quad\Longrightarrow\quad \tan 2\theta_m = \frac{\sin 2\theta}{\cos 2\theta - 2EV/\Delta m^2}.$$

Δm^2 is the difference of the squared masses, E the neutrino energy, θ the mixing angle in vacuum and V the potential of the electron neutrino in matter.

5.5 Row 6, COUPLED_SOLUTIONS_PROVENANCE: The mode equation of pair creation

The two-component amplitude $c(\tau)$ of a mode of dirac16complex in the Einstein–Kasner solution of the earliest regime obeys

$$i\,\frac{dc}{d\tau} = \omega\,(\cos\theta\,\sigma_1 + \sin\theta\,\sigma_2)\,c, \qquad \omega = K\sqrt{\tau^{-2p_1} + \tau^{-2q_1}}, \qquad \tan\theta = \tau^{p_1 - q_1}.$$

τ is the time since the singularity, K the scaling variable of the mode, $p_1 = (1+2\sqrt{2})/7$ and $q_1 = u_1 p_1$ with $u_1 = -1 + 1/\sqrt{2}$ the Kasner exponents of ordinary and auxiliary space, and σ_1, σ_2 Pauli matrices.

5.6 Row 7, COUPLED_SOLUTIONS: The minisuperspace action and the identities of its variation

For the metric $ds^2 = n^2 dx_0^2 - a^2(dx_1^2 + dx_2^2 + dx_3^2) - b^2(dx_4^2 + \cdots + dx_7^2)$, with EL_q the Euler–Lagrange expression for the variable q, evaluated at $n = 1$:

$$S = -\frac{1}{2\kappa}\int n\,a^3 b^4 \sum_k \alpha_k L_k\,dx_0 - \int nW\,dx_0,$$

$$\mathrm{EL}_n(na^3b^4L_k) = -2a^3b^4E^{(k)0}{}_0, \qquad \mathrm{EL}_a(na^3b^4L_k) = -6a^2b^4E^{(k)1}{}_1,$$

$$\mathrm{EL}_b(na^3b^4L_k) = -8a^3b^3E^{(k)4}{}_4.$$

$\kappa = 8\pi G_8$, α_k are the Lovelock couplings ($\alpha_0 = 2\Lambda$, $\alpha_1 = 1$), L_k the Lovelock densities, $E^{(k)A}{}_B$ the Lovelock tensors and $W(a,b)$ the comoving energy of the matter.

5.7 Row 8, COUPLED_SOLUTIONS_PROVENANCE: The comoving energy of one massless particle; the minisuperspace action and its identities

For one massless particle with comoving momenta k_a along ordinary space and k_b along the auxiliary space:

$$W = \omega = \sqrt{k_a^2/a^2 + k_b^2/b^2}, \qquad \partial_a W = -\frac{k_a^2}{a^3\omega}, \qquad \partial_b W = -\frac{k_b^2}{b^3\omega}.$$

$$S = -\frac{1}{2\kappa}\int n a^3 b^4 \sum_k \alpha_k L_k \, dx_0 - \int nW \, dx_0,$$

$$\mathrm{EL}_n(n a^3 b^4 L_k) = -2a^3 b^4 E^{(k)0}{}_0, \qquad \mathrm{EL}_a(n a^3 b^4 L_k) = -6a^2 b^4 E^{(k)1}{}_1,$$
$$\mathrm{EL}_b(n a^3 b^4 L_k) = -8a^3 b^3 E^{(k)4}{}_4.$$

The second display is the minisuperspace action with the identities of its variation, the same equations as those of the previous paragraph, except that the provenance file writes $n a^3 b^4$ without the thin space of $n\,a^3 b^4$; ω is the energy of the particle, and the derivatives $\partial_a W$, $\partial_b W$ enter the variations with respect to a and b.

5.8 Row 9, DARK_SECTOR: The Einstein-frame mass of a Kaluza–Klein level and its coupling to the radion

A level of Kaluza–Klein mass μ_n of dirac16complex (mass m) has, in the Einstein frame, the mass $M_E(\beta)$, with $M_n = M_E(0)$, and hence the coupling g_n to the canonical radion $\phi = \sqrt{12} M_{\mathrm{Pl}} \beta$:

$$M_E(\beta) = e^{-2\beta}\sqrt{m^2 + \mu_n^2 e^{-2\beta}}, \qquad g_n = \Big|\frac{dM_E}{d\phi}\Big|_{\beta=0} = \frac{2M_n + \mu_n^2/M_n}{\sqrt{12}\, M_{\mathrm{Pl}}}.$$

$\beta = \ln(b/b_0)$ is the radion and M_{Pl} the reduced Planck mass.

5.9 Row 10, DIRAC16COMPLEX: The five momenta of the exact mode sum

7+1.nb sums the mode algebra exactly, with symbolic $\vec{x}, \vec{y}$ and $m = 1$, for the symmetric set of five momenta

$$\vec{p} \in \big\{ \pm(2,2,4,0,0,0,0),\ \pm(0,0,0,2,2,0,0),\ \vec{0} \big\} \qquad (E = 5,5,3,3,1).$$

$E = \sqrt{m^2 + |\vec{p}|^2}$: $\sqrt{1+24} = 5$, $\sqrt{1+8} = 3$, $\sqrt{1} = 1$.

5.10 Row 11, DIRAC16COMPLEX_PROVENANCE: The orthogonal matrix U of the spinors

With the real spinors $u_s(\vec{p})$ of positive and $v_s(-\vec{p})$ of negative energy ($s = 1, \ldots, 8$, $u_s^{\mathsf{T}} u_s = 2E$) and the real symmetric plane-wave Dirac Hamiltonian $h(\vec{p})$:

$$U = \frac{1}{\sqrt{2E}}\big(u_1(\vec{p}), \ldots, u_8(\vec{p}), v_1(-\vec{p}), \ldots, v_8(-\vec{p})\big) \qquad \text{satisfies} \qquad U^{\mathsf{T}} h(\vec{p})\, U = \mathrm{diag}(E 1_8, -E 1_8).$$

U is real and orthogonal, $U^{\mathsf{T}} U = 1_{16}$; $h(\vec{p}) = p^i \alpha_i + m\Gamma^0$ with $\alpha_i = \Gamma^0\Gamma^i$ ($i = 1, \ldots, 7$, summed).

5.11 Row 14, KOHN_SHAM_GRAVITY_PROVENANCE: The completeness of the Θ_K and the Fierz identity

The 256 matrices Θ_K are complete (the first display), and this gives the Fierz identity (the second display) for commuting spinors a, b, c, d:

$$\sum_K (\Theta_K)_{cd} (\Theta^K)_{ab} = 16\, \delta_{ad}\delta_{bc},$$

$$\sum_{K\in r} (a\Theta_K b)(c\Theta^K d) = \sum_J C_{rJ}\, (a\Theta_J d)(c\Theta^J b), \qquad C_{rJ} = \frac{1}{256}\sum_{K\in r} \mathrm{tr}\big(\Theta_K \Theta_J \Theta^K \Theta^J\big).$$

Here $(aMb) = \sum_{i,j} a_i M_{ij} b_j$ for a 16×16 matrix M; $K \in r$ runs over the multi-indices of rank r and J over all 256; as subscripts in the first display, a, b, c, d are spinor indices $1, \ldots, 16$, and δ is the Kronecker delta.

5.12 Row 15, README_PROVENANCE: The notation of sections 7–10

With Ψ the field dirac16complex:

$$\Gamma_A = \Gamma_A^{(1,7)}, \qquad \Gamma^A = \eta_{(1,7)}^{AA}\Gamma_A, \qquad \Gamma^0 = \Gamma_0, \qquad \chi = \chi^{(1,7)} = i\Gamma_8^{(1,7)},$$
$$\bar{\Psi} = \Psi^\dagger\Gamma^0, \qquad \vec{x} = (x_1, \ldots, x_7), \qquad \vec{p} = (p^1, \ldots, p^7).$$

$\Gamma_A^{(1,7)}$ are the eight real 16×16 Dirac matrices of 7+1.nb, $\eta_{(1,7)} = \mathrm{diag}(+1, -1, \ldots, -1)$, and $\Gamma_8^{(1,7)} = \Gamma_0^{(1,7)} \cdots \Gamma_7^{(1,7)}$.

5.13 Row 16, STANDARD_MODEL: The electroweak scale and the masses at tree level

The reworded sentence states:

$$v = (\sqrt{2}G_F)^{-1/2} = 246.2196402 \text{ GeV}, \qquad m_W = \frac{gv}{2}, \qquad m_Z = \frac{\sqrt{g^2 + g'^2}\, v}{2}, \qquad m_H^2 = 2\lambda v^2.$$

G_F is the Fermi constant, g and g' the gauge couplings of $SU(2)_L$ and $U(1)_Y$, λ the quartic coupling of the Higgs field; the document writes the three masses inline, as $m_W = gv/2$ and $m_Z = \sqrt{g^2 + g'^2}\, v/2$.

5.14 Row 17, STANDARD_MODEL_SYMBOLS: The matrix notation

The reworded convention:

$$1_n = (\delta_{ij})_{i,j=1}^n, \qquad (M^\dagger)_{ij} = \overline{M_{ji}}, \qquad (M^*)_{ij} = \overline{M_{ij}},$$
$$(A \otimes B)_{(i-1)p+k,\,(j-1)q+l} = A_{ij}B_{kl} \quad (B \text{ of size } p \times q).$$

1_n is the $n \times n$ identity, $M^\dagger$ the conjugate transpose, M^* the complex conjugate (the bar is complex conjugation of a number) and $A \otimes B$ the Kronecker product, in Mathematica `KroneckerProduct[A, B]`; the document states these in words, the display writes them out.

6. The checks

6.1 The twelve check programs

The sheet Checks, first table: the outputs of the programs in the rebuild of a fresh clone of 102f399 (section 2.4 defines the counts).

program	what it checks	exit code	True	False	verdict
`tools/verify_7p1_nb.py`	The evaluated 7+1.nb: every check output, the printed numbers against the Python programs, the earlier sections against their backups	0	54	0	All cross-checks passed
`tools/ref44.py`	Rebuilds the objects of 4+4.nb	0	10	0	All printed results True
`tools/ref71.py`	Rebuilds the octonions and the Clifford matrices of 7+1.nb	0	25	0	All printed results True
`tools/extra71.py`	Further identities: boosts, rotations, the classification of the nine signatures, real solutions	0	23	0	All printed results True
`tools/ref_dirac16.py`	Sections 7–9: dirac16complex, Kohn–Sham, chiral Kohn–Sham	0	90	0	All checks passed
`tools/ref_gravity.py`	Sections 11–14: gravity	0	101	0	All checks passed
`tools/ref_ks_gravity.py`	Sections 15–16: Kohn–Sham theory in gravitational fields	0	29	0	All checks passed
`tools/ref_sm.py`	Sections 17–19: the Standard Model	0	33	0	All checks passed

program	what it checks	exit code	True	False	verdict
`tools/ref_coupled.py`	Sections 20–23: the coupled field equations and their solutions	0	50	0	All checks passed
`tools/ref_dark.py`	Section 24: the dark sector	0	20	0	All checks passed
`tools/ref_torus_brane.py`	Section 25: the auxiliary torus and the world brane	0	38	0	All checks passed
`tools/ref_radion.py`	Section 26: the radion	0	31	0	All checks passed
Total (12 programs)	**12 of 12 programs exit with code 0**		**504**	**0**	

6.2 The notebook 7+1.nb

The notebook contains 479 checks `check["name", test]`: 443 print their result where they stand, and 36 are made by the function `frameChecks` of section 13.1 of the notebook (nine for each of four metrics) and printed by its summary table. The notebook was evaluated in a fresh kernel and finished on 2026-10-28 08:57:47 PDT.

quantity	value
checks in the notebook	479
checks that returned True	479
… printed where they stand	443
… made by `frameChecks` (section 13.1), printed by the summary table	36
message (error) cells	0
input cells	189
output cells	554
evaluation time (seconds)	283.883
Wolfram version	15.0.1 for Microsoft Windows (64-bit), fresh kernel

6.3 A fresh clone of 102f399, rebuilt from scratch

The repository was cloned from GitHub into an empty directory, and every document was rebuilt there with `tools/make_docs.py` (which also runs the twelve programs); the rebuilt files were then compared with the committed ones — Markdown and LaTeX byte for byte, PDF by its text (`pdftotext`) and its page count (`pdfinfo`), since a PDF also records the time of its creation.

quantity	value
files tracked in the repository	184
Markdown and LaTeX files compared	60
… identical byte for byte	60
PDFs compared	30
… with the same text and page count	30
documents with layout problems (loose or overflowing lines, errors)	0
other files changed by the rebuild	0

7. The private files

The sheet Private files: files of the author that are in the working folder of the project but are not published. Their names are not given here.

file	what it is	on GitHub?	how it is kept out
The radion note (PDF)	A PDF printout of the author's e-mail of 28 October 2026 on the radion	No	Listed in `.git/info/exclude` on the author's computer
The vielbein notes (PDF)	A PDF printout of the author's private notes on vielbeins in n dimensions	No	Listed in `.git/info/exclude` on the author's computer
A third private PDF	Private working material of the author	No	Listed in `.gitignore`
`.gitignore`	The ignore rules of the repository	No	Lists itself

`.gitignore` lists the files that Git does not track in every clone of the repository; `.git/info/exclude` lists files that Git does not track in one working folder only, and is itself never published. The check: `git ls-files` of the fresh clone of 102f399 lists none of these files.

8. The formulas of the spreadsheet

8.1 Cells, ranges and functions

A cell is named by its column letter and its row number (`B5`: column B, row 5); `Commits!E6` is the cell E6 of the sheet Commits, and a sheet name with a space is quoted (`'Private files'!C2`). A range `A2:A5` is the rectangle of cells from A2 to A5. With R a range, $v(c)$ the value of the cell c, x a value and y a value, the functions and operators of the spreadsheet are

$$\mathrm{SUM}(R) = \sum_{c\in R,\ v(c)\text{ a number}} v(c), \qquad \mathrm{COUNTA}(R) = \#\{c \in R :\ v(c) \text{ not empty}\},$$

$$\mathrm{COUNTIF}(R, x) = \#\{c \in R :\ v(c) = x\} \quad \text{(texts compared without regard to case)},$$

$$x + y = \text{the sum}, \qquad x \,\&\, y = \text{the text of } x \text{ followed by the text of } y.$$

8.2 The 35 formulas and their values

In the order of the sheets. The values are computed here, from the data file, by an evaluator of these functions; the same values are stored in the spreadsheet.

sheet	cell	formula	value
Summary	B5	`=Commits!A5`	102f399
Summary	B6	`=COUNTA(Commits!A2:A5)`	4
Summary	B7	`=Commits!E6`	206
Summary	B8	`=COUNTA(Documents!A2:A31)`	30
Summary	B9	`=Documents!D32`	478
Summary	B10	`=Documents!E32`	40
Summary	B11	`=Documents!F32`	12
Summary	B12	`=Documents!G32`	0
Summary	B13	`=Documents!H32`	0
Summary	B14	`=Documents!I32`	0
Summary	B15	`=COUNTIF(Fixes!D2:D19,"Display")+COUNTIF(Fixes!D2:D19,"Code block")+COUNTIF(Fixes!D2:D19,"Reworded")`	16
Summary	B16	`=COUNTA(Checks!A2:A13)`	12
Summary	B17	`=COUNTIF(Checks!C2:C13,0)`	12
Summary	B18	`=Checks!D14`	504
Summary	B19	`=Checks!E14`	0
Summary	B20	`=Checks!B17`	479
Summary	B21	`=Checks!B18`	479

sheet	cell	formula	value
Summary	B22	`=Checks!B21`	0
Summary	B23	`=Checks!B30&" of "&Checks!B29`	60 of 60
Summary	B24	`=Checks!B32&" of "&Checks!B31`	30 of 30
Summary	B25	`=COUNTIF('Private files'!C2:C5,"Yes")`	0
Commits	E6	`=SUM(E2:E5)`	206
Commits	F6	`=SUM(F2:F5)`	25,646
Commits	G6	`=SUM(G2:G5)`	5,357
Documents	D32	`=SUM(D2:D31)`	478
Documents	E32	`=SUM(E2:E31)`	40
Documents	F32	`=SUM(F2:F31)`	12
Documents	G32	`=SUM(G2:G31)`	0
Documents	H32	`=SUM(H2:H31)`	0
Documents	I32	`=SUM(I2:I31)`	0
Documents	J32	`=COUNTIF(J2:J31,"Yes")`	11
Documents	K32	`=COUNTIF(K2:K31,"Yes")`	9
Checks	B14	`=COUNTIF(C2:C13,0)&" of 12 programs exit with code 0"`	12 of 12 programs exit with code 0
Checks	D14	`=SUM(D2:D13)`	504
Checks	E14	`=SUM(E2:E13)`	0

Check. The spreadsheet stores, for each formula, the value that LibreOffice computed when it saved the file; 35 of the 35 stored values equal the values of the table.

9. How to reproduce

The spreadsheet. The command below writes `7plus1_status_2026-10-28.xlsx` from `tools/status_2026-10-28.json` with openpyxl; if LibreOffice is installed, it then opens and saves the file in headless mode, which computes and stores the value of every formula, and it checks that no formula gives an error:

```
python3 tools/make_status_xlsx.py
```

The documents. This document, its symbol dictionary and its provenance record (Markdown, LaTeX, PDF); the command runs the twelve check programs first:

```
python3 tools/make_docs.py 7plus1_status_2026-10-28 \
    7plus1_status_2026-10-28_SYMBOLS 7plus1_status_2026-10-28_PROVENANCE
```

The numbers of the data file come from the repository — the commits (section 3), the changed documents and the pages:

```
git show --shortstat --format= <commit>
git show --name-only --format= <commit>
git show 102f399:<document>.pdf > x.pdf && pdfinfo x.pdf
```

— and from the logs of LaTeX in `build/docs` (the loose lines), `build/docs/runs.json` (the outputs of the programs), the evaluation record `tools/eval_log_7p1.txt` and the evaluated 7+1.nb (the notebook), and a fresh clone (section 6.3).